

WHIMSY

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1 Dyadic drift reformulation of the boundary substitution problem

We work with the parity-core pair

$$E = \frac{1}{8}(1, 0, 6, 0, 1), \quad O = \frac{1}{2}(0, 1, 0, 1, 0),$$

so that

$$X \in \{0, 2, 4\}, \quad Y \in \{1, 3\}, \quad \mathbb{E}X = \mathbb{E}Y = 2.$$

The purpose of this section is to replace the fixed-height core–tail compression by a genuinely multiscale formulation. The basic object is no longer a packet cut from a remote critical tail, but a history-dependent redistribution of the local reserve drift across dyadic reserve bands.

The resulting formulation has three exact ingredients:

1. an exact parametrization of every phase-resolved bulk fan by two positive submeasures;
2. an exact simple-symmetric bulk walk, hence an exact gambler’s-ruin factor $1/2$ between consecutive dyadic scales;
3. a single remaining positive prefix-causal regeneration problem at the hard boundary.

1.1 The exact bulk coupling

Consider the coupling

$$\kappa(0, 1) = \frac{1}{8}, \quad \kappa(2, 1) = \frac{3}{8}, \quad \kappa(2, 3) = \frac{3}{8}, \quad \kappa(4, 3) = \frac{1}{8},$$

with all other entries equal to zero.

Its row law is E , its column law is O , and every occupied cell satisfies

$$X \leq Y + 1.$$

If the current reserve is $m + J$, with $m \geq 1$, then the residual reserve after one step is

$$m + J + Y - X.$$

Writing U for the upward shift operator, the three public children are therefore

$$\begin{aligned} T_0 &= U^{m+1}T, \\ T_2 &= \frac{1}{2}U^{m-1}T + \frac{1}{2}U^{m+1}T, \\ T_4 &= U^{m-1}T. \end{aligned}$$

In particular, every state at distance $m \geq 2$ from the hard boundary is closed exactly by this bulk coupling. The only unresolved layer is $m = 1$.

1.2 Phase-resolved parametrization by two submeasures

Let

$$B = \sum_{j \geq 0} b_j \delta_j$$

be the current reserve law.

For each $j \geq 0$, choose two numbers

$$0 \leq u_j, v_j \leq \frac{b_j}{2}.$$

Interpret them through the joint law

$$\Pr(J = j, X = 0, Y = 1) = u_j, \quad (1)$$

$$\Pr(J = j, X = 2, Y = 1) = \frac{b_j}{2} - u_j, \quad (2)$$

$$\Pr(J = j, X = 2, Y = 3) = \frac{b_j}{2} - v_j, \quad (3)$$

$$\Pr(J = j, X = 4, Y = 3) = v_j. \quad (4)$$

For every fixed reserve state j , the Y -marginal is exactly

$$\Pr(Y = 1 \mid J = j) = \Pr(Y = 3 \mid J = j) = \frac{1}{2}.$$

Hence the fresh-input law O is preserved pointwise in the hidden state.

The public output law is exactly E if and only if

$$\boxed{\sum_{j \geq 0} u_j = \frac{1}{8}, \quad \sum_{j \geq 0} v_j = \frac{1}{8}.} \quad (\text{DD1})$$

At the hard boundary $j = 0$, nonnegativity of reserve rules out $(X, Y) = (2, 1)$ and $(4, 3)$. Therefore necessarily

$$\boxed{u_0 = \frac{b_0}{2}, \quad v_0 = 0.} \quad (\text{DD2})$$

In particular,

$$b_0 \leq \frac{1}{4}.$$

The three active child laws are then explicit.

[Exact drift-coordinate fan] For every admissible pair u, v satisfying (??)–(??), the active children are

$$\boxed{B_0(k) = 8u_{k-1}, \quad k \geq 1,} \quad (\text{DD3})$$

with $B_0(0) = 0$,

$$\boxed{B_4(k) = 8v_{k+1}, \quad k \geq 0,} \quad (\text{DD4})$$

and

$$\boxed{B_2(k) = \frac{4}{3} \left(\frac{b_{k+1}}{2} - u_{k+1} + \frac{b_{k-1}}{2} - v_{k-1} \right),} \quad (\text{DD5})$$

where all terms with negative indices are interpreted as zero.

Dūkaz. The formulas follow by conditioning on X and using

$$J' = J + Y - X.$$

For $X = 0$, necessarily $Y = 1$, hence $J' = J + 1$. Since $\Pr(X = 0) = 1/8$, division by $1/8$ gives (??).

For $X = 4$, necessarily $Y = 3$, hence $J' = J - 1$. Division by $\Pr(X = 4) = 1/8$ gives (??).

For $X = 2$, the two possibilities are $Y = 1$ and $Y = 3$, producing increments -1 and $+1$. Division by $\Pr(X = 2) = 3/4$ gives (??). $\square$

Thus the local causal freedom is completely represented by the pair of positive submeasures u, v .

1.3 The drift ledger

For $b_j > 0$, the conditional probabilities of moving one unit upward or downward are

$$p_j^+ = \frac{1}{2} + \frac{u_j - v_j}{b_j},$$

and

$$p_j^- = \frac{1}{2} - \frac{u_j - v_j}{b_j}.$$

Hence

$$\mathbb{E}[\Delta J \mid J = j] = 2 \frac{u_j - v_j}{b_j}. \quad (\text{DD6})$$

Since the total masses of u and v are equal,

$$\sum_j (u_j - v_j) = 0.$$

Therefore

$$\sum_j b_j \mathbb{E}[\Delta J \mid J = j] = 0. \quad (\text{DD7})$$

At the boundary, (??) gives

$$\mathbb{E}[\Delta J \mid J = 0] = 1. \quad (\text{DD8})$$

Consequently the positive reflection drift at the hard boundary must be paid exactly by negative drift in the positive reserve states:

$$\sum_{j \geq 1} (u_j - v_j) = -\frac{b_0}{2}. \quad (\text{DD9})$$

Equation (??) is the fundamental conservation law of the multiscale problem. Boundary repair does not consume a separate finite fuel reservoir; instead it redistributes drift from higher reserve bands toward the hard boundary.

1.4 Exact bulk random walk

Forget temporarily the phase dependence of u, v and follow one tagged piece of bulk phase under the exact coupling κ .

If its current distance from the hard boundary is $m \geq 2$, then after averaging over the public output $X \sim E$,

$$\frac{1}{8} \delta_{m+1} + \frac{6}{8} \left(\frac{1}{2} \delta_{m-1} + \frac{1}{2} \delta_{m+1} \right) + \frac{1}{8} \delta_{m-1} = \frac{1}{2} \delta_{m-1} + \frac{1}{2} \delta_{m+1}.$$

Thus the tagged bulk phase performs the simple symmetric random walk

$$S_{t+1} = S_t \pm 1$$

with equal probabilities until it hits the boundary layer.

Lemma 1 (Exact dyadic gambler's ruin). *Let $L \geq 1$, and let S_t be the simple symmetric walk on $\{0, 1, \dots, 2L\}$ started from $S_0 = L$. Let*

$$\tau = \inf\{t : S_t \in \{0, 2L\}\}.$$

Then

$$\Pr_L(S_\tau = 2L) = \frac{1}{2}, \quad \Pr_L(S_\tau = 0) = \frac{1}{2}. \quad (\text{DD10})$$

Důkaz. The stopped walk $S_{t \wedge \tau}$ is a bounded martingale. Hence

$$L = \mathbb{E}_L S_\tau = 2L \Pr_L(S_\tau = 2L),$$

which gives (??). □

The Green function of the same killed walk is explicit.

Lemma 2 (Exact occupation kernel). *For $1 \leq r \leq 2L - 1$,*

$$G_L(r) := \mathbb{E}_L \left[\sum_{t < \tau} \mathbf{1}_{\{S_t = r\}} \right]$$

is

$$G_L(r) = \begin{cases} r, & 1 \leq r \leq L, \\ 2L - r, & L \leq r \leq 2L - 1. \end{cases} \quad (\text{DD11})$$

In particular,

$$G_L(1) = 1 \quad (\text{DD12})$$

and

$$\mathbb{E}_L \tau = L^2. \quad (\text{DD13})$$

Thus one unit of phase mass injected at scale L creates exactly one expected visit to the layer adjacent to the boundary before the excursion terminates at either 0 or $2L$.

At the same time exactly one half of the phase mass exits through the upper dyadic boundary $2L$.

This is the intrinsic origin of the coefficient $1/2$.

1.5 Why a finite fuel reservoir cannot close the problem

A fixed-height decomposition separates the law into a finite head and a single remote critical tail. Such a representation cannot exploit (??): finite mass which diffuses upward from the head cannot be reclassified positively as new leading $1/j$ tail mass.

Consequently the remote tail behaves as a nonrenewable external fuel reservoir.

This is not the correct physical picture of the exact bulk dynamics. Under (??)–(??), reserve mass continuously moves between neighboring heights, and the same mass may subsequently return to the boundary or escape into a higher dyadic band.

The correct conserved resource is therefore not “tail mass” but the multiscale drift flux

$$u - v.$$

A particularly instructive extreme boundary repair is the complete 3:1 exchange: one unit of boundary mass is pushed upward while three units of donor mass pay the corresponding negative drift.

If R denotes the available donor-to-boundary mass ratio, repeated 04-cycling gives

$$R_{n+1} = \frac{R_n - 3}{4}. \quad (\text{DD14})$$

Hence every finite donor reservoir is exhausted after finitely many such cycles. The successful mechanism must therefore distribute the donor debt over unboundedly many reserve scales and replenish high-scale mass by bulk diffusion.

1.6 Dyadic drift packets

Let

$$L_s := 2^s, \quad s = 0, 1, 2, \dots$$

We seek a decomposition

$$\boxed{u - v = \sum_{s \geq 0} \Delta_s} \tag{DD15}$$

into signed scale packets satisfying

$$\boxed{\sum_{j \geq 0} \Delta_s(j) = 0.} \tag{DD16}$$

The positive part of Δ_s supplies upward reflection near the hard boundary, while the negative part is supported at reserve heights of order L_s .

The bulk dynamics then transports the corresponding phase packet through the dyadic interval

$$[L_s, 2L_s].$$

By Lemma 1, precisely one half of the tagged phase mass reaches the next scale:

$$\boxed{m_{s+1} = \frac{1}{2}m_s} \tag{DD17}$$

for an ideal scale packet.

By Lemma 2, its total expected boundary-layer work before exiting the scale is of order m_s , with an exact unit occupation factor at distance one.

Thus

$$m_s \asymp 2^{-s}, \quad L_s \asymp 2^s,$$

and consequently

$$m_s L_s = O(1).$$

This is exactly the scale balance required for logarithmic total reserve cost.

1.7 The new load-bearing lemma

The preceding identities are exact. What remains is a positivity and prefix-compatibility statement.

[Dyadic drift regeneration lemma] There exists a history-indexed family of positive reserve laws

$$\mathcal{V}$$

containing the critical root, together with a decomposition into dyadic phase packets

$$\Phi_s, \quad s \geq 0,$$

and admissible history-dependent controls

$$u_h, v_h,$$

with the following properties.

At every reachable history h :

1. The controls satisfy

$$0 \leq u_h(j), v_h(j) \leq \frac{1}{2}B_h(j),$$

$$\sum_j u_h(j) = \sum_j v_h(j) = \frac{1}{8},$$

and at the hard boundary

$$u_h(0) = \frac{1}{2}B_h(0), \quad v_h(0) = 0.$$

2. The three public children defined by (??)–(??) again belong to $\mathcal{V}$.
3. The drift field admits a positive multiscale decomposition

$$u_h - v_h = \sum_{s \geq 0} \Delta_{h,s},$$

where every $\Delta_{h,s}$ has zero total mass and its negative part is supported in a reserve band

$$c_1 2^s \leq j \leq c_2 2^s$$

for absolute constants $0 < c_1 < c_2 < \infty$.

4. The phase associated with scale s evolves in the positive-distance bulk according to the exact symmetric walk of Lemma 1. Its upper exit from the dyadic band becomes the next-scale phase.
5. The continuing phase mass obeys the exact regeneration law

$$\boxed{|\Phi_{s+1}| = \frac{1}{2} |\Phi_s|.} \tag{DD18}$$

6. The lower-exit mass is not discarded. It is absorbed into admissible boundary/core states and is re-emitted by the same controls u_h, v_h without creating a new uncompensated drift debt at the same scale.
7. Every scale packet obeys

$$|\Phi_s| \leq C 2^{-s}, \quad J \leq K 2^s$$

on its support, for absolute constants $C, K < \infty$.

The distinction in item 6 is essential.

The coefficient $1/2$ in (??) is not interpreted as killing half of the phase. The lower half is recycled through the boundary system, whereas the upper half advances to the next dyadic scale.

This prevents the construction from collapsing into a finite-fuel architecture.

1.8 Relation to the original Lemma S

The original boundary substitution problem asks for a positive history-indexed phase whose surviving distinguished component advances one scale and has coefficient exactly $1/2$.

Lemma 1.7 supplies precisely such a mechanism.

Indeed, inside every positive-distance band the exact bulk coupling gives a simple symmetric walk. Starting from the middle of the dyadic band, Lemma 1 sends exactly one half of the phase mass through the upper boundary.

The lower half returns to the boundary system and is recycled by the same prefix-causal controls.

Thus the coefficient

$$\frac{1}{2}$$

is no longer a parameter imposed on a packet decomposition: it is generated intrinsically by the exact bulk dynamics.

Consequently a proof of Lemma 1.7, together with finite exact entry from the critical root into the dyadic grammar, would imply the boundary substitution Lemma S and hence

$$\boxed{\mathcal{B}_n = O(\log n).}$$

Combined with the known logarithmic lower bound, this would yield

$$\boxed{\mathcal{B}_n = \Theta(\log n).}$$

1.9 Equivalent solver target

The remaining problem is therefore no longer a search for a fixed packet policy.

It is the following scale-homogeneous positive flow problem:

Construct history-dependent positive controls u, v satisfying the exact fan equations (??)–(??) such that the boundary reflection debt

$$\frac{1}{2}B_h(0)$$

is distributed over dyadic reserve bands by zero-mass drift packets $\Delta_{h,s}$, while the exact symmetric bulk walk regenerates the negative donor flux and transfers exactly one half of every distinguished phase from scale 2^s to scale 2^{s+1} .

Equivalently, one seeks a positive scale-homogeneous substitution system whose transient bulk kernel is the killed simple symmetric walk and whose boundary operator closes the returned mass without creating stationary scale probability.

This is the proposed solver-level reformulation of Lemma S.

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All identities preceding Hypothesis 1.7 are exact. The sole remaining load-bearing statement is the positive, prefix-compatible multiscale closure in Lemma 1.7.